

GENERAL SPECIFICATION G2.27

Training, Advisory and Technical Support Plan

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G2.27 TRAINING, ADVISORY AND TECHNICAL SUPPORT PLAN

G2.27.1 Training Requirements

Training requirements shall include:

- (a) Training shall be provided on Site or at the premises of specialist vendors in Singapore as applicable and shall commence when directed by the S.O.. At Site, training shall take place in the S.O.'s office. For off-Site training the Contractor shall provide a training venue. Wherever training is carried out, the Contractor shall provide all necessary training materials and all audio-visual equipment necessary to conduct the training.
- (b) The Contractor shall coordinate closely with the Board and the S.O. with regard to the training programme. Where required by the S.O., the Contractor shall re-schedule the training sessions to prevent clashes in timing.
- (c) Training shall empower the Board's personnel to perform optimum operation of equipment, routine and periodic maintenance of every item of equipment, ensure health and safety at work, security and as may be otherwise required for satisfactory operation and maintenance of the equipment.
- (d) The training shall comprise two main elements:
 - (i) Classroom and practical training in groups. This includes the video recording of the classroom and practical training. Three copies of the video recordings should be handed over to the Board in the form of digital video discs (DVDs). The training shall include PLC control logic training for each system, with details related to the alarms connected with relevant instruments, the use and explanation of all interlocks, and address any queries by any staff of the Board. All instrument ranges and specifications should be detailed with the training; and
 - (ii) "Hands-on" training while the Board's operation and maintenance staff are operating the Works and carrying out routine maintenance under the direction and supervision of the Contractor during Testing & Commissioning, Plant Performance Testing and any interim operations period.
- (e) During the classroom and practical training, the Board will make available selected staff for the Contractor to train. The Contractor shall allow for comprehensive training in at least four separate sessions for each of the following disciplines:
 - (i) Management staff;
 - (ii) Operation staff including chemists and laboratory personnel;
 - (iii) Electrical and ICA maintenance staff; and
 - (iv) Mechanical maintenance staff).
- (f) The training programme shall be designed to accommodate the Board's shift patterns (24 hours, 3 shifts per day).
- (g) The exact number of personnel attending the training sessions will be finalised prior to the training.

- (h) During the hands-on training, the Contractor shall remain fully responsible for the operation and maintenance of the plant, but shall utilise the Board's staff to the fullest extent possible, commensurate with their proficiency at the time, with a view to providing a smooth transition of responsibility for operation and routine maintenance from the Contractor to the Board.
- (i) The Contractor shall adjust the training schedule for interruptions in operability of equipment and process units.
- (j) Training shall be performed by specialist personnel specifically allocated to the task. Such training shall be specific to the trainees' needs and shall be appropriate to staff with the basic skills related to their grade.
- (k) The Contractor shall adjust the training outline to ensure training of appropriate personnel as deemed necessary by the Board, and to allow full participation of the Board's personnel by the manufacturers' representatives.
- (l) The Contractor shall ensure that training on individual items of plant or equipment shall be carried out by the manufacturer or specialist sub-contractor. Provision shall be made for both classroom and hands-on-training in this respect. The Contractor shall be responsible for ensuring that such training is properly coordinated into the overall programme, and that the training is of the type, quality and standard specified.
- (m) The authorized representative (trainer) of the manufacturer shall be factory trained, and experienced in the technical applications, installation, operation, and maintenance of respective equipment, subsystem, or system, with full authority by the equipment manufacturer to issue the certifications required of the manufacturer. Additional qualifications may be specified elsewhere.
- (n) The Contractor shall submit details of the number of training specialists he shall assign to each phase of training together with details of their duties, qualifications, capabilities and experience to the S.O. for approval. Manufacturer's representative shall be familiar with facility operation and maintenance requirements as well as with specified equipment.
- (o) The Contractor shall maintain a register of attendance at training sessions and the subjects covered.
- (p) Training shall be based on the relevant operating and maintenance manuals which shall be available before training commences.
 - (i) Detailed description, including operation principles/mechanism and demonstration on operation and control of all electrical, mechanical and ICA equipment.

G2.27.2 Training Methodology

The training methodology shall consider:

- (a) The method of delivery needs be structured to include theory, group discussion and part practical elements supported by audio visual aids, handouts and demonstrations. Handouts shall include extracts from operating and maintenance manuals supplemented by specially prepared material and shall be prepared so as to be collated to form a set of course notes for use by attendees for future reference. Audio-visual aids shall include relevant video presentations prepared by plant and equipment manufacturers, as well as those prepared during plant installation and testing together with samples of the equipment to be installed on site; and
- (b) Three hard copies, plus three soft copies on DVD with searchable function, of the course notes and trainers' presentation materials shall be submitted to the S.O. a minimum of three weeks prior to each training session. Videos to be used as part of the training shall also be included in the DVDs.

G2.27.3 Training Programme

The training programme shall include:

- (a) Classroom Training shall start at an appropriate time during the installation, commissioning or testing of equipment and shall be completed before the assets or Plant is ready to be handed over to the Board. Following the classroom training, the Contractor shall provide "hands-on" training and full direction and supervision of the Board's staff.
- (b) The Contractor shall continue to train the Board's personnel to enhance their skills in the operation and maintenance of the Plant in the months following the completion of interim operations period of the Plant by the Contractor, up to the completion of DLP.
- (c) The extent/duration of training provided shall also take into account the number of Board's personnel involved, shift patterns and scope of training as specified elsewhere:
 - (i) The Contractor shall allow for adequate time (not necessarily consecutive days) to allow at least one rotation through the shift schedule to complete each training session for each shift; and
 - (ii) The Board shall reserve the right to change the Training Schedule submitted for review to ensure that the Board's staff is available and all shift personnel are able to attend.
- (d) To ensure a conducive learning environment, the number of trainees per theory and group discussion class shall not exceed ten and the number of trainees per practical and/or "hands on" training session shall not exceed five, unless otherwise directed by S.O.. If the number of trainees exceeds the specified class size, the Contractor shall conduct additional sessions to accommodate these trainees.
- (e) Each individual training class shall not exceed 4 hours. A mid-session break shall be provided for all training classes.

G2.27.4 Training Material

The training material shall include:

- (a) A comprehensive overall training manual covering all disciplines, which shall be submitted to the S.O. for approval 3 months before the commencement of the training programme. Subsequently the approved manual shall be submitted at least one month before the commencement of training in the following format:
 - (i) Three hard copies and one soft copy on DVD with searchable function shall be submitted to the DDC; and
 - (ii) The presentation requirements of the manual shall be the same as those for the operating and maintenance manual. The soft copy of the manual shall be in the form a single document with interactive links allowing easy cross-referencing. The training manual shall include all relevant text, training presentations and technical brochures required to form a complete document. The manual shall be updated as necessary throughout the training programme.
- (b) In addition, each trainee shall receive one set of bound documents including the following:
 - (i) Course notes – course description and outline of course content;
 - (ii) Estimated training times for each topic;
 - (iii) Name and resume of the instructors providing the training; and
 - (iv) Training information from the overall training manual appropriate to their particular skill areas (e.g. operator, electrical and mechanical maintenance).
- (c) The training material shall be in a user-friendly style and shall cover in general terms the main technical procedures and activities associated with the Plant or assets. The documents shall be in a format suitable for use in future training of Board's personnel up to operating shift supervisor level.
- (d) Training topics shall include as a minimum:
 - (i) Overview of the equipment provided under the Contract and the duty requirements;
 - (ii) Principles of operation of equipment;
 - (iii) Detailed description of equipment; A detailed description of all equipment and components supplied under the contract including their function;
 - (iv) Detailed operating procedures;
 - (v) Troubleshooting and fault finding;
 - (vi) Maintenance procedures; from day to day maintenance up to major refurbishment;
 - (vii) Familiarisation in the field; a physical walk around the installed equipment with demonstrations of points discussed in previous sections;
 - (viii) Spare parts; a listing of recommended spare parts and their availability; and
 - (ix) General Q&A.

- (e) The training material shall contain Illustrations with PowerPoint/ animations/ 3D models, schematics of every process involved in operation and maintenance of the Plant.
- (f) Training documents shall be securely bound in such a manner that individual pages can be added or replaced so that the documents can be kept up to date.
- (g) On completion of the training programme, the training manual and other documentation, updated to take account of changes during the performance of the programme. Two copies of the updated manual shall be submitted to the S.O. in PDF format on DVD format.
- (h) The Contractor shall also carry out video recording of the training sessions and submit edited video footage in an approved format to the S.O. of all supplied training.

G2.27.5 Training Assessment

The Contactor shall provide a training assessment for each trainee. This shall include:

- (a) The training shall be monitored continuously by the Contractor jointly with the S.O. and the Board's representative. In order to achieve an acceptable standard of competence. The Contractor shall build into the training an assessment monitoring programme as set out below or a comparable assessment method, an example of a training assessment form is attached in Appendix 1, approved by the DDC:
 - (i) Each session shall include small groups of trainees being given a task/scenario/questionnaire relevant to the equipment which will require them to use the knowledge learnt during the session to feedback and demonstrate understanding in such a manner as to promote trainee participation without putting individuals under pressure; and
 - (ii) This type of assessment shall be adopted for both theoretical and practical sessions and time shall be built into the course programme for this purpose.
- (b) Training shall be considered satisfactory if at least seventy-five (75) per cent of the Board's personnel trained consider the training adequate and this is reflected in the training feedback forms. Retraining shall be done as many times as necessary until the training is considered satisfactory. All retraining shall be done at the Contractor's expense.

G2.27.6 Training Records

- (a) The Contractor shall maintain a complete record of training and testing on video on a format to be submitted and approved by the S.O. and at the end submit to the S.O. a master copy, and to the Board four copies on DVD.
- (b) The contractor shall complete a training form that contains, but not limited to, the following information:
 - (i) The name and or number of the training session;
 - (ii) The equipment or process covered in the training session;
 - (iii) Date of training;
 - (iv) Relevant section in the O&M manual, if applicable;

- (v) Attendees; and
- (vi) Training Type (Classroom, Hands-on, Specialist etc.).
- (c) An example of a training session record form is attached as an Appendix 2.

G2.27.7 Technical Support

- (a) Throughout the DLP, the Contractor shall advise the Board on the proper operation of the equipment and provide technical support to the Board to overcome operating and maintenance difficulties.
- (b) The Contractor shall attend to all requests for advice and assistance, both oral and written, made by the Board or the S.O. and shall keep a written record of these requests with a copy submitted to the S.O. monthly.
- (c) Any advice or technical support given by the Contractor to the Board, as a result of requests from them, and the action taken by them as a result of such technical support shall be recorded by the Contractor and copies shall be submitted to the S.O. monthly.

G2.27.8 Appendices

- (a) APPENDIX G2.27A – Operations Training Programme Example.
- (b) APPENDIX G2.27B – Training Session Record Form Example.
- (c) APPENDIX G2.27C – Training Assessment Form Example.

END OF SECTION

Appendix G2.27A – Operations Training Programme Example

Target Start Date	No.	Session Description	Provider	Target Audience	Duration (Hours)	Average Score	Date completed	REMARKS

Appendix G2.27B – Training Session Record Form Example

Training Session Number			
Equipment or Process covered in training			
Date of training			
Relevant section in O&M manual if applicable			
Trainer(s)			
Attendees			
Training Type (Classroom, Hands-on, Specialist etc.)			
Comments			
Training outlines met?			
<i>Verified By</i> PUB Plant Operations Manager		<i>Date:</i>	
<i>Verified By</i> S.O. Interface Manager		<i>Date:</i>	

Appendix G2.27C – Training Assessment Form Example

Training Session Number									
Equipment or Process Covered in Training									
Date of Training									
Objective									
Please complete the following questions. Indicate your response with a X. 10 = Very Good 5 = Average 1 = Very Poor									
1. Were the defined outlines of the training adequately achieved?									
10	9	8	7	6	5	4	3	2	1
2. Was the presentation material adequate?									
10	9	8	7	6	5	4	3	2	1
3. Was the Training presented in a clear and concise manner?									
10	9	8	7	6	5	4	3	2	1
4. Was the timing of the Training appropriate?									
10	9	8	7	6	5	4	3	2	1
5. Are you confident you could start to Operate/Maintain this equipment or process based on the training received?									
10	9	8	7	6	5	4	3	2	1
Could anything be done to improve this training in future?									
Additional Comments									
Verified By PUB Plant Operations Manager						Date:			
Verified By S.O.Interface Manager						Date:			